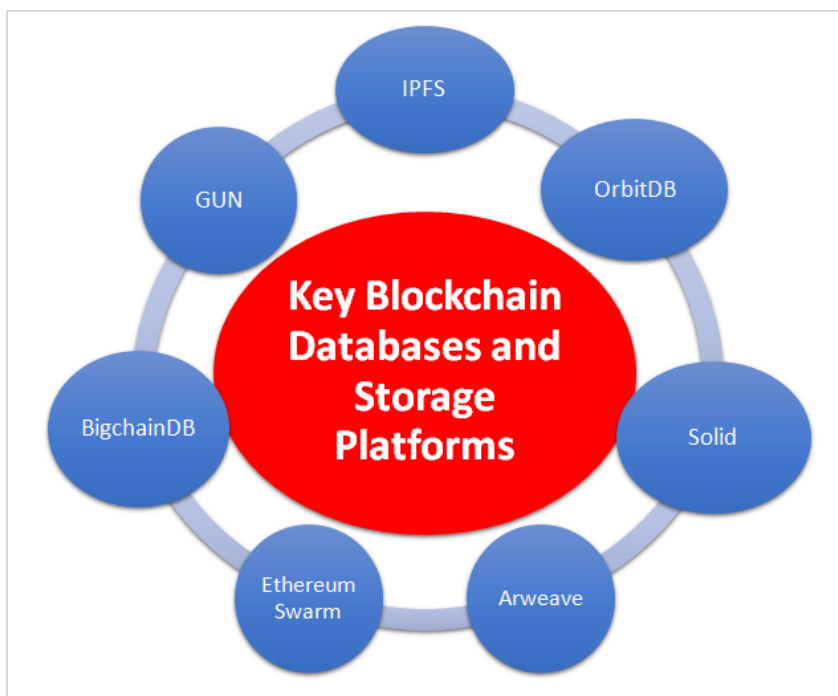




operations to the database layer, enabling faster transaction processing and improved scalability. For example, some blockchain platforms employ databases like RocksDB or Cassandra to handle the storage and retrieval of transaction data, enhancing overall network performance.

Free and open source databases and storage platforms for blockchain development

When it comes to developing decentralised applications (dApps), databases play a crucial role in storing and managing data in a decentralised manner. Here are some notable databases suitable for decentralised app development, along with their URLs and descriptions.

IPFS (InterPlanetary File System) -<https://ipfs.io/>

IPFS is a distributed file system that provides decentralised storage and retrieval of files. It uses content-addressing to uniquely identify files and leverages a network of nodes to store and distribute data. IPFS

is often used in conjunction with blockchain technology to store large files, multimedia content, or extensive metadata in a decentralised and censorship-resistant manner.

OrbitDB -<https://orbitdb.org/>

OrbitDB is a distributed peer-to-peer database built on top of IPFS. It offers a simple API for creating and managing databases that can be shared and synchronised across multiple nodes. OrbitDB supports different database types such as key-value stores, document stores, and event logs, making it suitable for various decentralised app use cases.

Ethereum Swarm

-<https://swarm.ethereum.org/>

Ethereum Swarm is a decentralised storage and content distribution platform that is part of the Ethereum ecosystem. It allows developers to store and retrieve data in a decentralised manner using a distributed network of nodes. Swarm integrates seamlessly with Ethereum smart contracts, enabling dApp developers to store and access data securely and efficiently.

GUN (graph database)

-<https://gun.eco/>

GUN is a decentralised graph database that enables real-time data synchronisation across peers. It provides a distributed, offline-first data storage and synchronisation mechanism, making it ideal for building decentralised applications that require real-time collaboration and data consistency. GUN's data synchronisation capabilities make it suitable for applications like messaging, collaborative editing, and social networks.

Holochain -<https://holochain.org/>

Holochain is a framework for building decentralised applications that can be run on peer-to-peer networks. It provides a distributed hash table (DHT) that allows developers to store and retrieve data in a decentralised manner. Holochain focuses on agent-centric development, where each user has their own chain of data that they control, enhancing privacy and scalability in dApp development.

Arweave -<https://www.arweave.org/>

Arweave is a permanent and decentralised storage network that provides immutable and censorship-resistant data storage. It utilises a blockchain-like structure called the 'blockweave' to store data and ensures its availability over a long period of time. Arweave is often used for decentralised app development, particularly for applications that require permanent storage of critical data or content.

BigchainDB

-<https://www.bigchaindb.com/>

BigchainDB combines the benefits of distributed databases and blockchain technology to provide a scalable and decentralised data storage solution. It offers high throughput and low latency, making it suitable for decentralised applications that require fast and efficient data management. BigchainDB allows developers to store, retrieve, and query